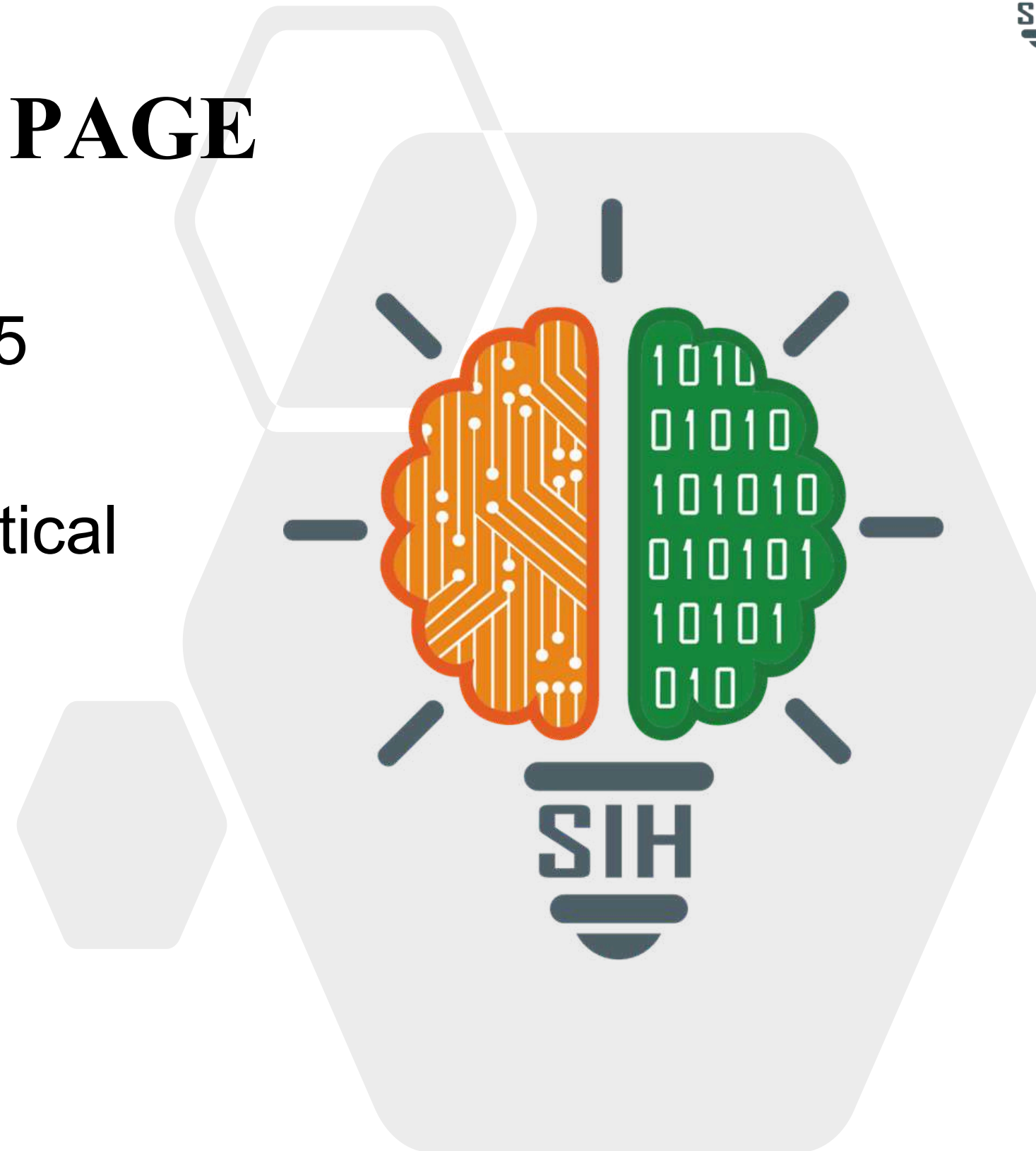


TITLE PAGE

- **Problem Statement ID –** SIH26185
- **Problem Statement Title-** Helmet mounted conformal antenna for tactical communications in urban CQB environments
- **Theme-** Robotics and Drones
- **PS Category-** Hardware
- **Team ID-**
- **Team Name (Registered on portal)**

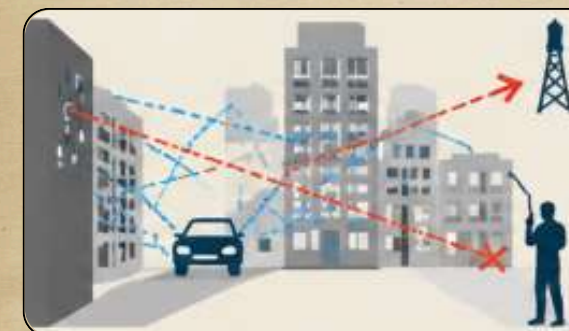




Helmet mounted conformal antenna for tactical communications in urban CQB environments

THE PROBLEM

URBAN ENVIRONMENTS CREATE DIFFICULT RF CONDITIONS:



- Concrete walls and buildings attenuate RF signals
- Reflections and diffraction create multiple signal paths
- Multipath fading causes unstable signal strength
- Vehicles and other structures can obstruct communication
- Communication reliability can vary with user movement and orientation

Conventional antennas are not optimized for wearable integration.



Conventional external antennas are inconvenient for wearable systems.

- Protruding antennas increase snagging and physical-damage concerns
- Antenna placement may vary with equipment configuration
- Conventional designs may not account for helmet curvature
- Helmet and body proximity can alter antenna performance

Our Approach



A compact, low-profile antenna designed to conform to the helmet surface and characterized under realistic conditions.

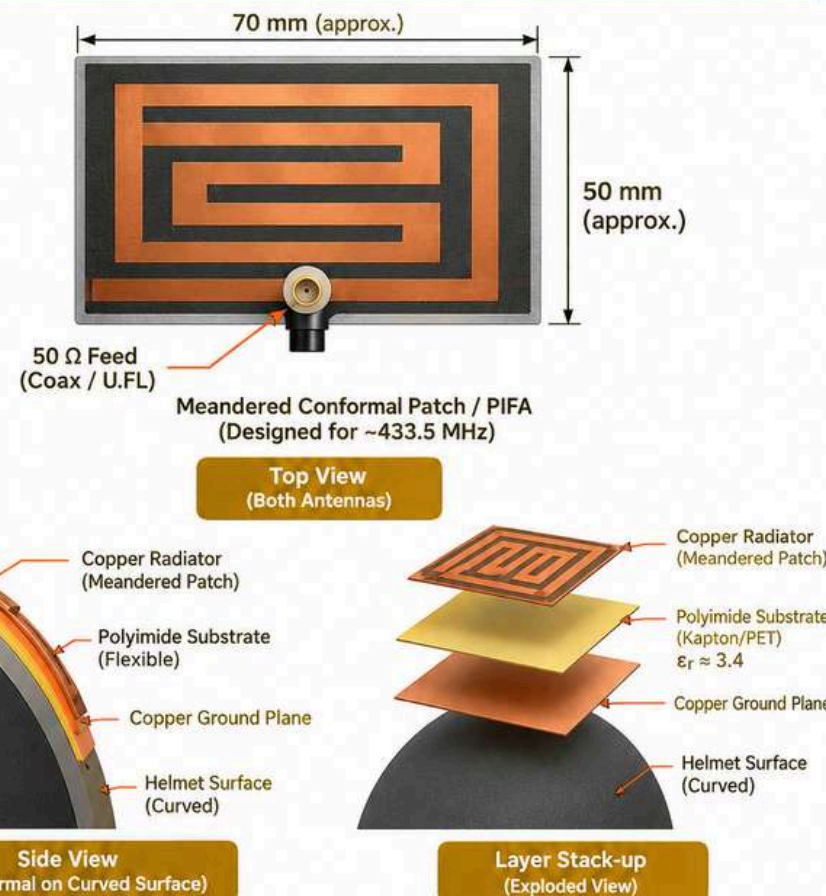
THE NEED





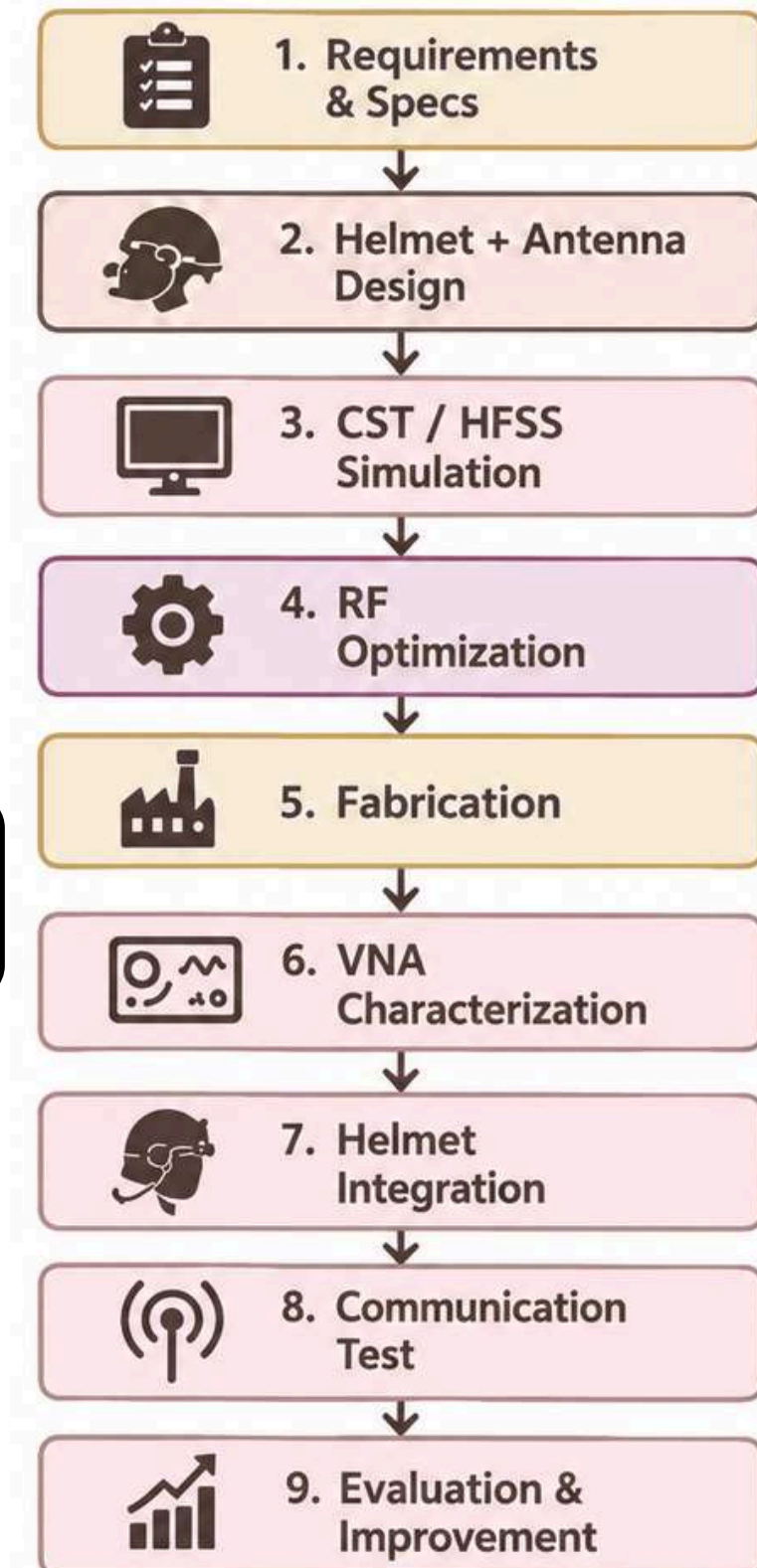
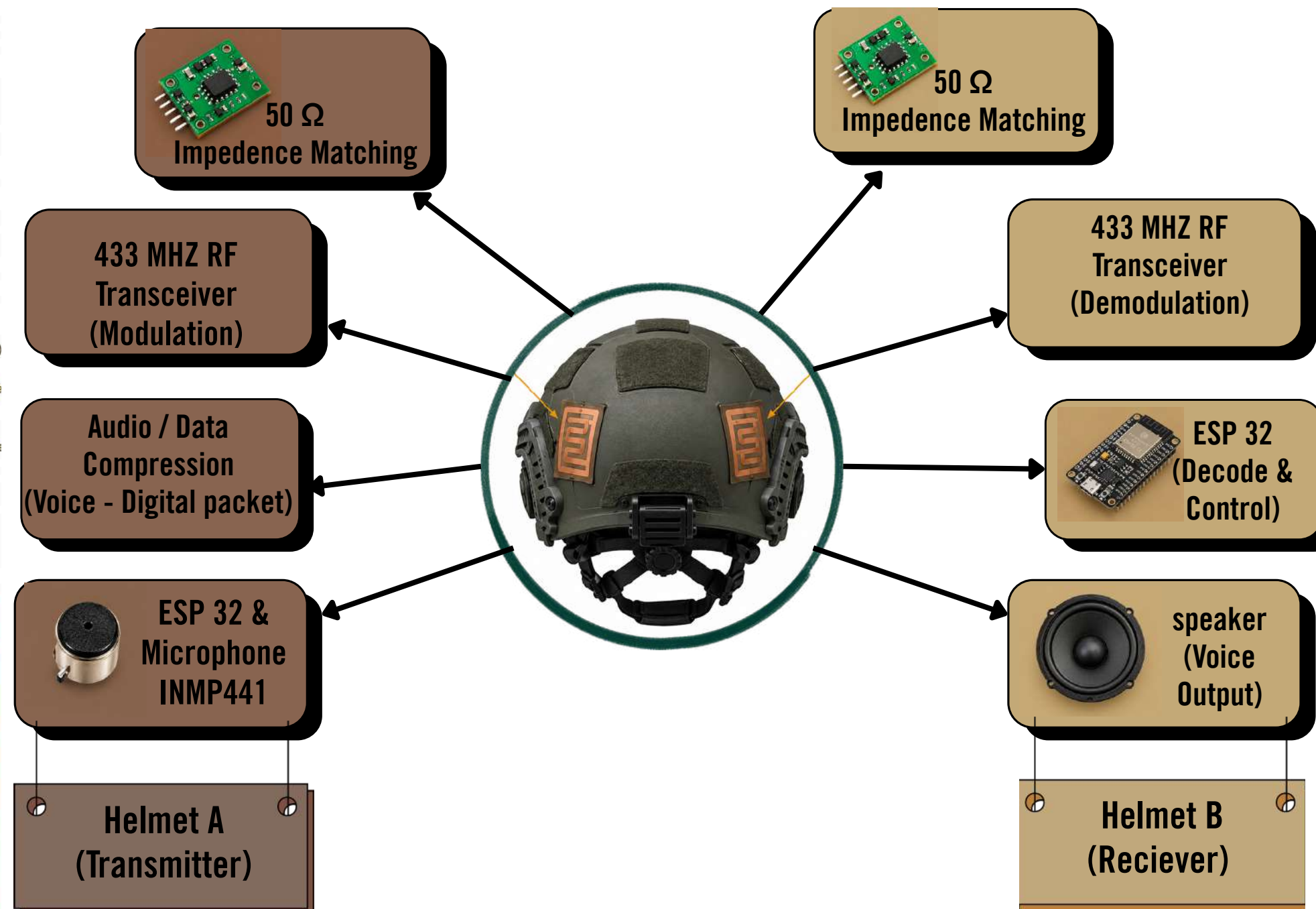
TECHNICAL APPROACH

Antenna Design



Design Specifications

Parameter	Specification
Design frequency	~ 433.5 MHz
Antenna type	Meandered conformal microstrip patch / PIFA
Feed impedance	50 ohm
Substrate	Flexible PCB (Polyimide/Kapton), PET, or FR-4 (for early testing)
Radiator	Copper foil or PCB copper layer
Ground plane	Copper layer behind the substrate
Target S11	< -10 dB near 433 MHz
Target VSWR	< 2:1





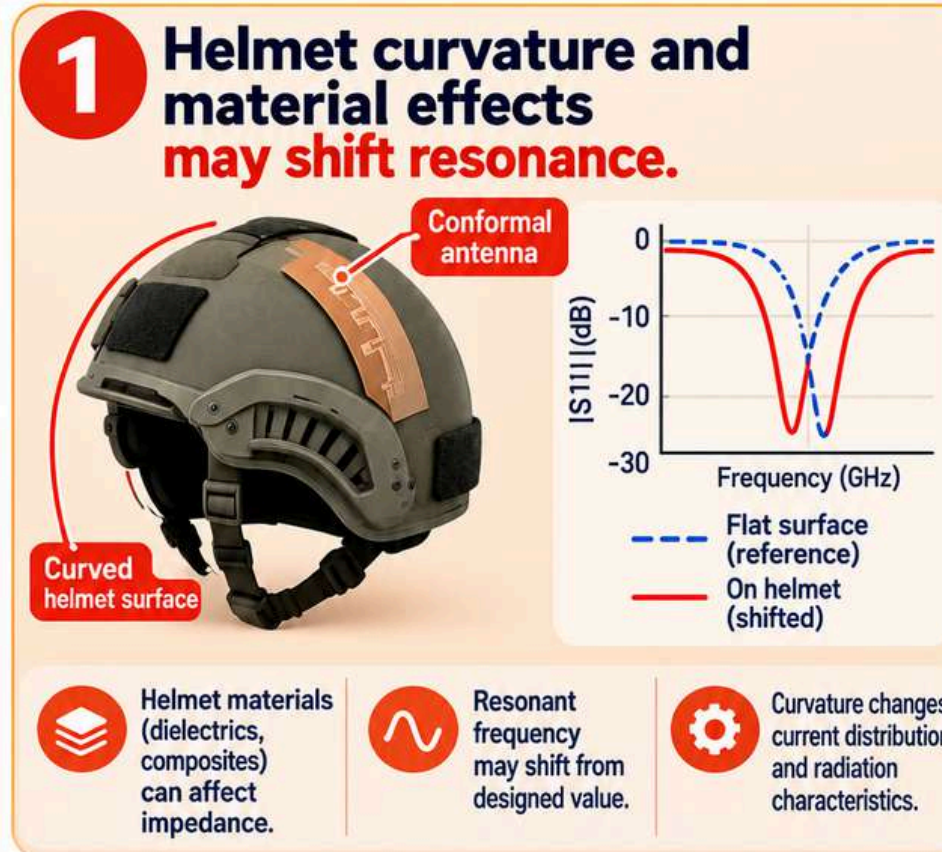
FEASIBILITY AND VIABILITY



FEASIBILITY

- Compact conformal antenna can be integrated with helmet curvature.
- CST/HFSS enables EM simulation and optimization.
- Prototype can be fabricated and validated using a VNA.
- Controlled **low-power RF** communication testing is achievable.

CHALLENGES & RISKS



MITIGATION STRATEGIES

- **Model helmet** and relevant materials in CST/HFSS.
- Optimize antenna geometry iteratively.
- Perform **VNA measurements** before and after helmet integration.
- Test different orientations and representative obstacles.





IMPACT AND BENEFITS



IMPACT



- **Technical Impact** – Develops a compact wearable RF antenna platform.
- **Communication Impact** – Enables study of connectivity under urban multipath and obstruction.
- **Economic Impact** – Supports potential low-cost, locally developed RF hardware.
- **Technology Impact** – Builds capability in antenna design, RF engineering and wearable electronics.

IMPACT



Improves communication in challenging environments.



Supports real-world testing and validation.



Enables compact and wearable integration.



Provides a scalable platform for future enhancements.

BENEFITS



Small & Lightweight
Easy to integrate with a helmet.



Comfortable
No long antenna sticking out.



Better Integration
Follows the curved helmet surface.



Easy to Improve
Can be tested and optimized for better performance.

WEARABLE | RELIABLE | FUTURE READY



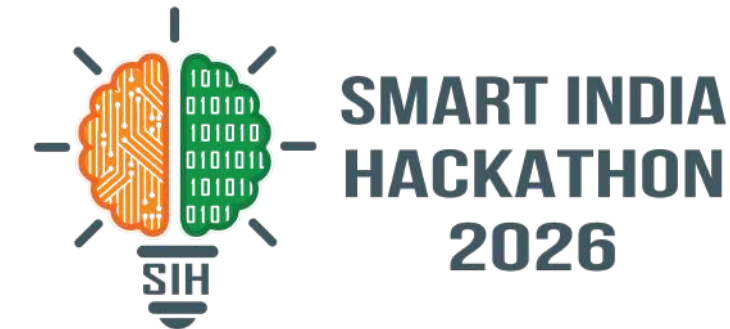
BENEFITS



- **Low-Profile Design** – Integrates with the helmet without protruding structures.
- **Wearer-Friendly** – Lightweight and suitable for wearable integration.
- **Reliable Connectivity** – Supports communication in challenging urban environments.
- **Scalable Platform** – Can be extended to dual-antenna, diversity and future wearable RF systems.



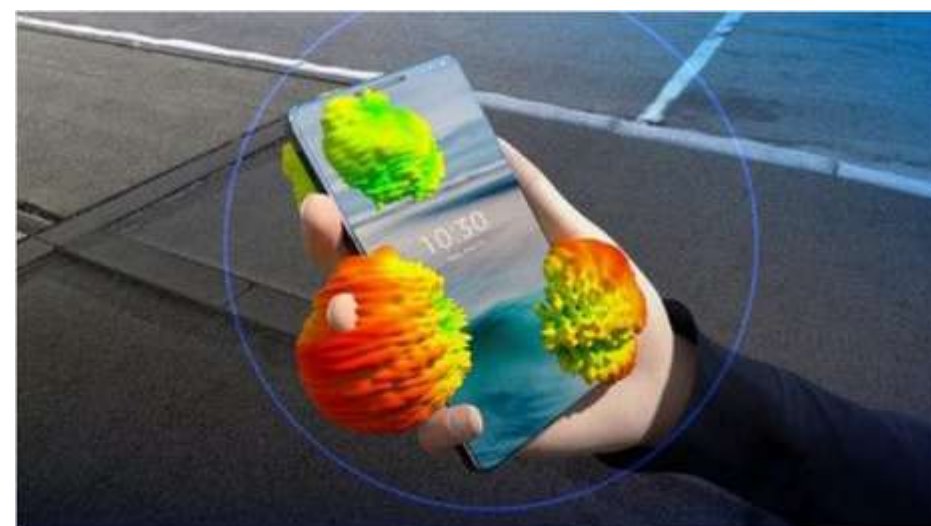
RESEARCH AND REFERENCES



REFERENCES & RESEARCH WORK

- Wearable & Conformal Antennas – Design challenges and body-loading effects.
- CST / Ansys HFSS – Antenna simulation and optimization.
- VNA Testing – S11, VSWR, bandwidth and resonance validation.
- Research Focus – Helmet curvature → Body loading → Simulation → Fabrication → Testing.

USEFUL LINKS



Antenna Simulation With CST Studio Suite | Dassault Systèmes®

Watch this webinar to discover how electromagnetic simulation with SIMULIA CST Studio Suite is used by industry-leading companies to develop and optimize antenna systems.

MEGATrends



Fundamentals of Antenna

Antennas are one of the important elements for any wireless communication links. They have several characteristics which depend on the application type. For beginners, it is important to know the working methods of an antenna in the wire...

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Antenna Design & Modeling Software | Ansys

Electromagnetic simulation of antenna design and its interaction with the entire system enables you to evaluate antenna placement, EMI/co-site interference and more.

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